

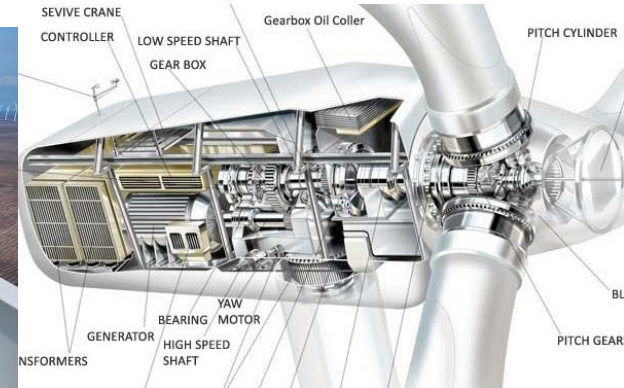
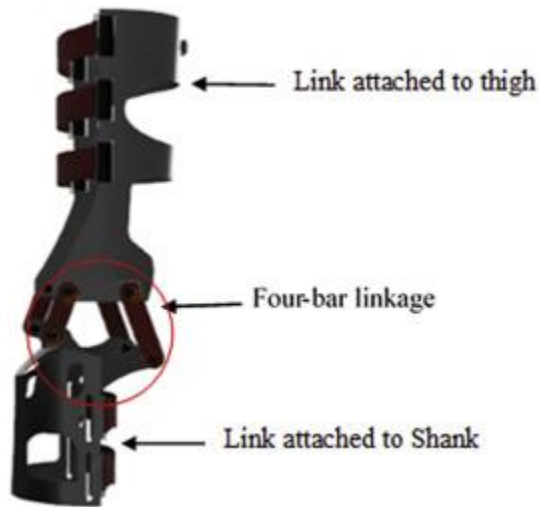
Machine Design

20 some years ago



Machine Design

Today...



Lecture Topics

Mechanisms

Linkages
Kinematic Diagrams
Mobility
Instant Centers
Mechanical Advantage
Graphical Synthesis
Analytical Synthesis

Machine Elements

Fasteners
Introduction to Transmissions
Transmission Ratios
Forces in Gears
Failure of Gears
Bearings

Mechatronics

Introduction to Mechatronics
Motors and Actuators
Analog Sensors
Digital Sensors
Signals
Control

Modern Machine Design

An Analogy¹...

- In Dynamics:
 - **Kinematics:** The study of motion without regard to forces.
 - **Kinetics:** The study of forces on systems in motion.

In Design:

Mechanism:

- Device which transforms motion to some desirable pattern and typically develops very low forces and transmits little power.
- A system of elements arranged to transmit motion in a predetermined fashion.

Examples: pencil sharpener, a camera shutter, an analog clock, a folding chair, an adjustable desk lamp, and an umbrella.

Machine:

- Typically contains *mechanisms* which are designed to provide significant forces and transmit significant power.
- A system of elements arranged to transmit motion and energy in a predetermined fashion.

Examples: food blender, a bank vault door, an automobile transmission, a bulldozer, a robot, and amusement park ride.

Machines are composed of Mechanisms

A **mechanism** is a device for transferring motion and/or force from a source to an output. In a mechanism, at least one link has been grounded, or attached to the frame of reference (which itself may be in motion.)

Landing Gear Mechanism



When it works...

<http://www.youtube.com/watch?v=aTjUFQvstjU>

<https://www.youtube.com/watch?v=xkp3XRlqAy0>

When it doesn't work...

<http://www.youtube.com/watch?v=ANLgWY6I9uk>

Types of Motion¹

- **Pure Rotation:** The body possesses one point (center of rotation) which has no motion with respect to the “stationary” frame of reference. All other points on the body describe arcs about that center. A reference line drawn on the body through the center changes only its angular orientation.
- **Pure Translation:** All points on the body describe parallel (curvilinear or rectilinear) paths. A reference line drawn on the body changes its linear position but does not change its angular orientation.
- **Complex motion:** A simultaneous combination of rotation and translation. Any reference line drawn on the body will change both its linear position and its angular orientation. Points on the body will travel nonparallel paths, and there will be, at every instant a center of rotation, which will continuously change location.

ME350 PUZZLE

What do the following items have in common?

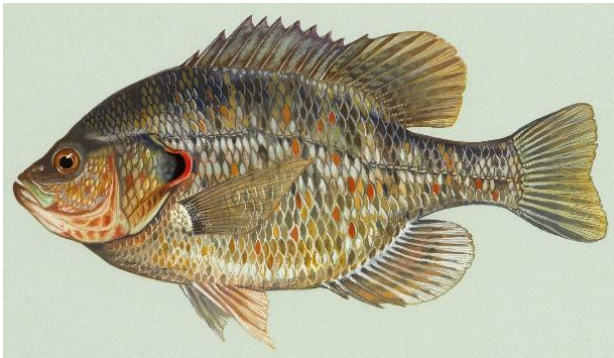
1) Vise-grip
Locking Pliers



2) Umbrella



3) Opercular Jaw



4) Automotive Suspension



ME350 PUZZLE

What do the following items have in common?

Answer: They all have linkages

1) Vise-grip
Locking Pliers

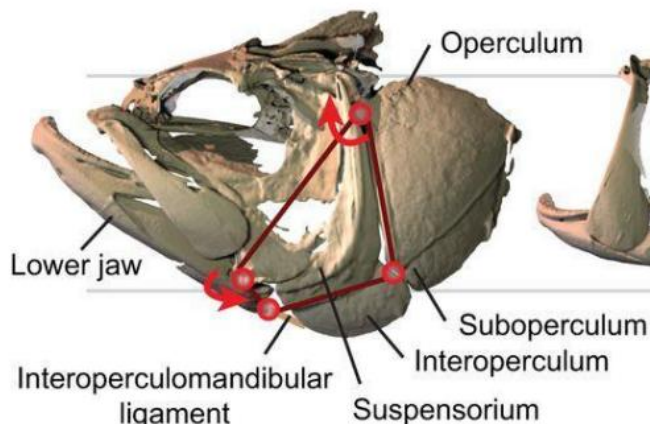


2) Umbrella

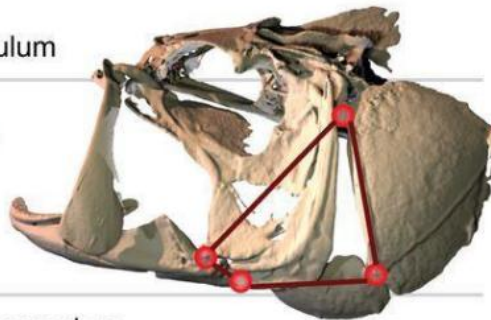


3) Opercular Jaw

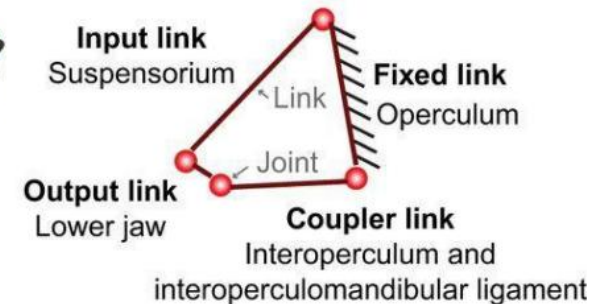
A



B

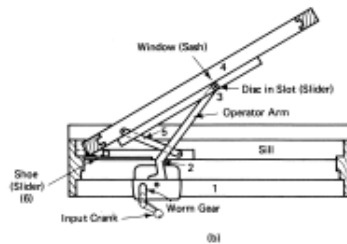
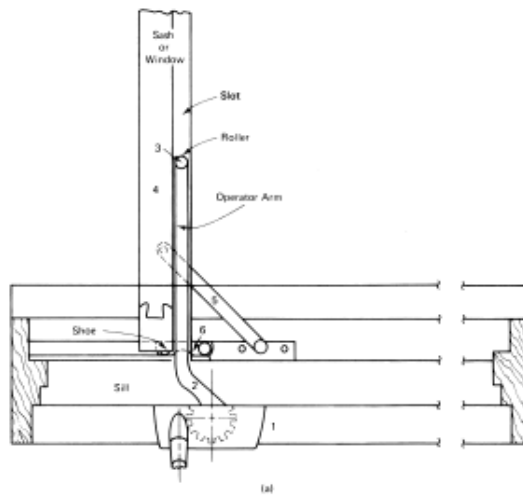


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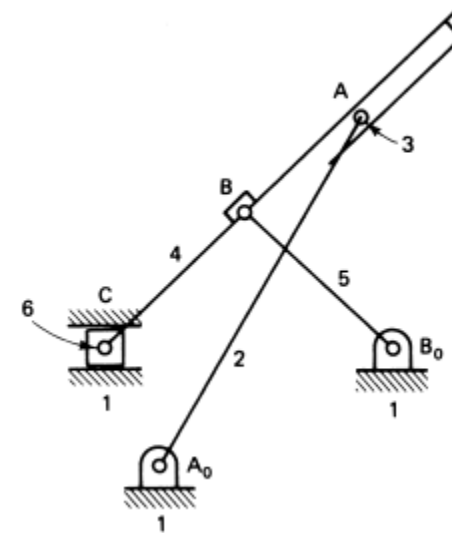


Kinematic Diagrams

We draw kinematic diagrams to reduce mechanisms into a simpler form so we can analyze them.

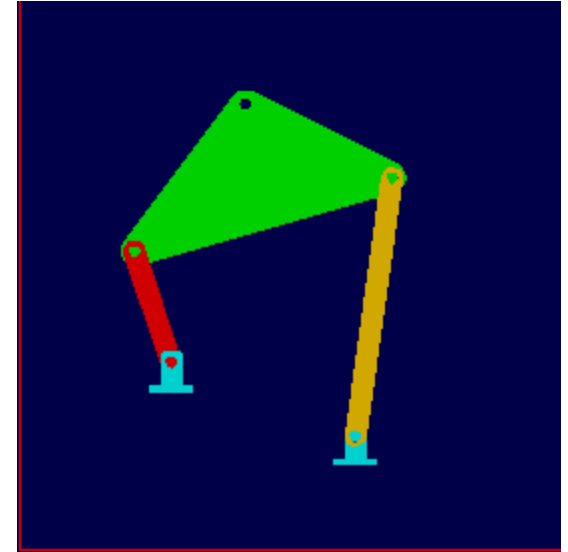
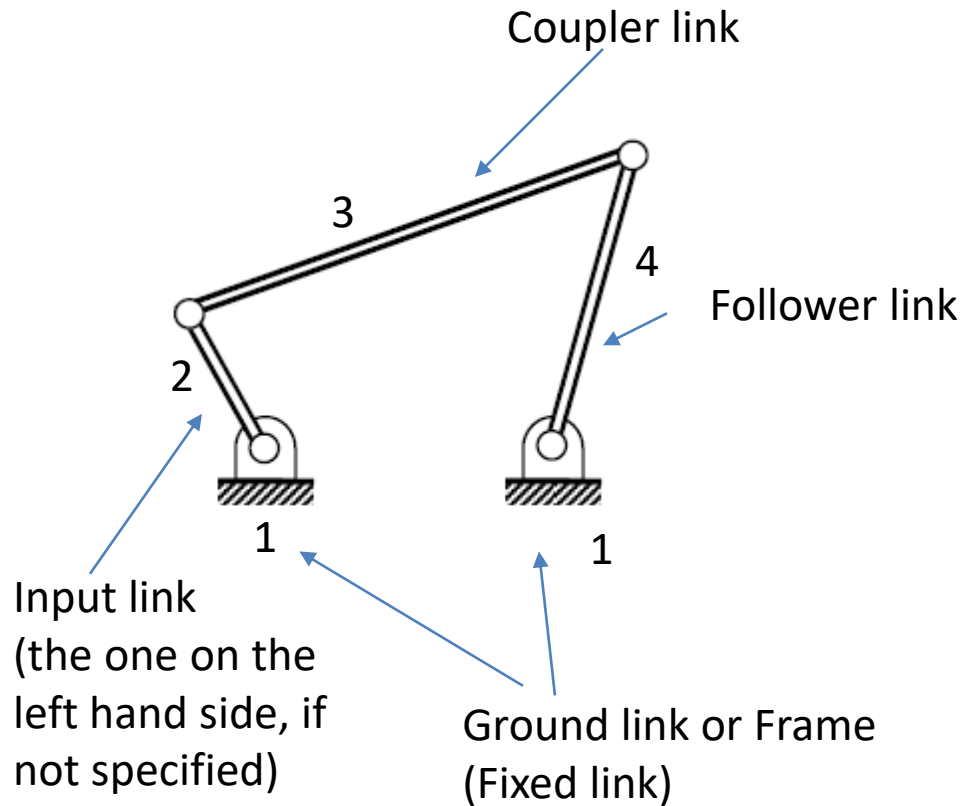


Casement Window Mechanism



Kinematic Diagram

Terminology of a Four-Bar Linkage



Always label ground link as “1”

Linkage

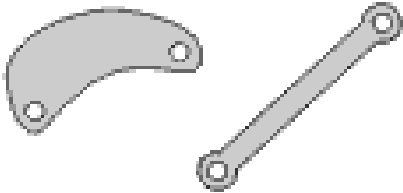
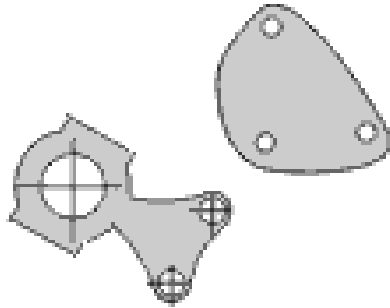
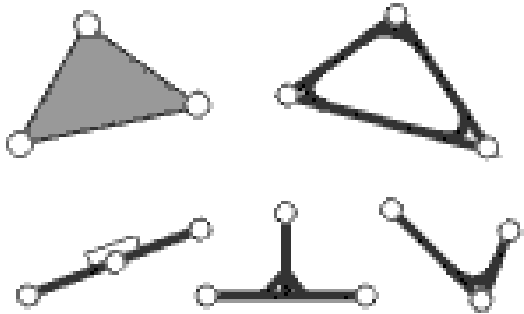
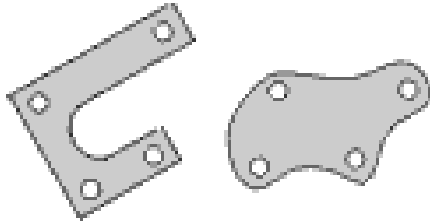
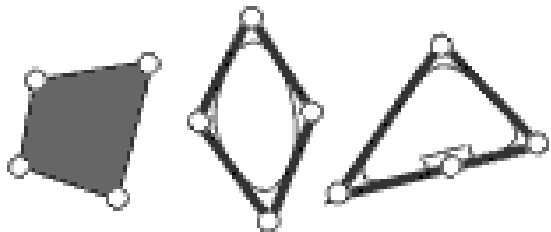
Linkage: an assembly of **links** connected by **joints**

Link: a rigid body with at least two joints

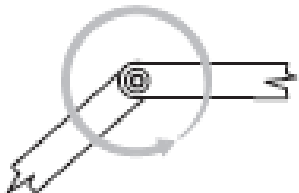
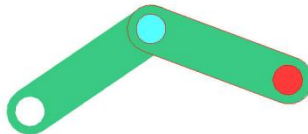
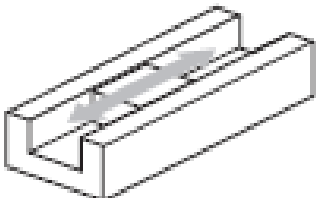
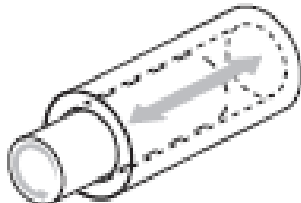
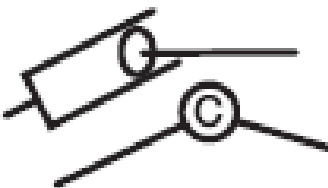
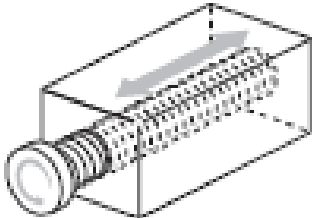
Joint: the connection between two links



Link: a rigid body with at least two joints (points of attachment to other links)

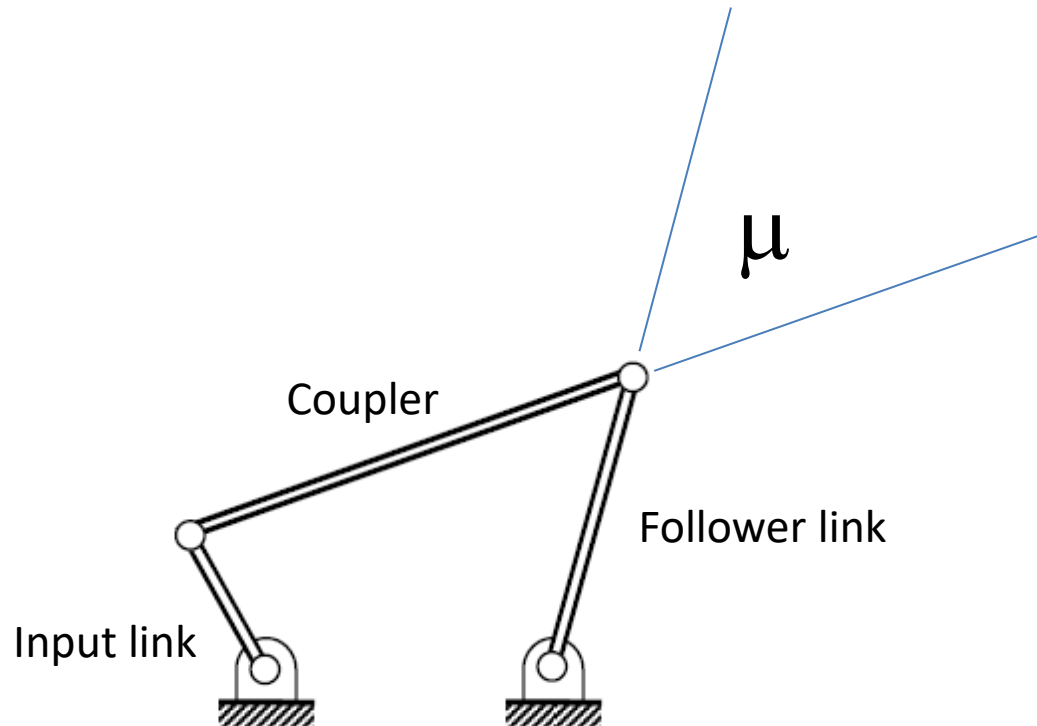
Link Types	Physical forms	Kinematic Representations
Binary		
Ternary		
Quaternary		

Some basic Types of Joints

Joint Type	Depiction	Schematic	Example	Degrees of Freedom
Pin Revolute				1 Rotational
Slider Prismatic				1 Translational
Slider Cylindrical				2 1-Rotational 1-Translational
Screw				1 1-Rotational or 1-Translational (not independent motions)

Transmission Angle, μ

The transmission angle is a means of determining the effectiveness of transferring motion and force to the output link.



It is defined as the acute angle between the coupler and the follower.

Poor Transmission Angle

- Ideal range of transmission angle
$$\mu \geq 30^\circ$$
$$(\mu \leq 150^\circ)$$
- The following may occur if transmission angle is poor:
 - Mechanism binds (no motion results)
 - Joints fail (pieces go flying)
 - Wrong motion (can't control which way it comes out of the toggle position)

Toggle positions

- Input link collinear with coupler
 - Infinite Mechanical Advantage (theoretically)
 - μ not necessarily = 0
- Coupler collinear with output link (follower)
 - “Change point” (dead center)
 - $\mu = 0$
 - Can’t control which way it comes out of the toggle position

